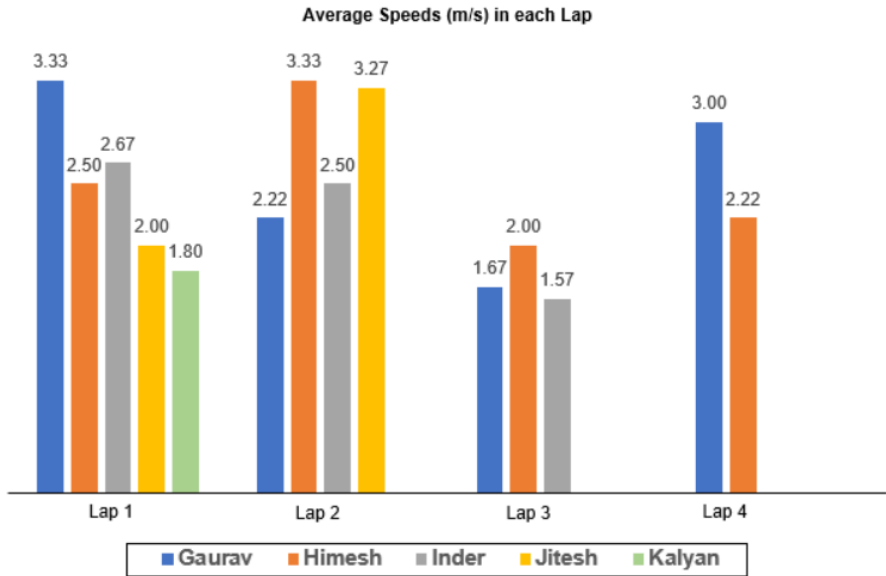


**DIRECTIONS** for questions 1 to 5: Answer these questions on the basis of the information given below.

Five students – Gaurav, Himesh, Inder, Jitesh and Kalyan – participated in a race in which the length of each lap was 400 m. In the race, in each lap, the student who is in the last position will be eliminated when the person just in front of him finishes that lap. For example, in the first lap, the student who is in the fifth position will be eliminated when the student in the fourth position finishes the first lap. The students in contention run the race continuously without any break until there is only one person remaining, who is then declared the winner of the race. The last person to be eliminated is declared the runner-up. The following bar graph presents the average speed (in meters per second, rounded off to two decimal places) of each student in each lap until they were eliminated:



**Q1. DIRECTIONS** *for question 1:* Select the correct alternative from the given choices.

The total distance (in meters) that Kalyan ran in the race was

- ☐ a) 111.11.
- ☒ b) 360. ✓ **Your answer is correct**
- ☐ c) 395.
- ☐ d) 180.

**Q2. DIRECTIONS** *for question 2:* Type in your answer in the input box provided below the question.

What was the total distance (rounded to the nearest meter) run by Jitesh in the race?

**Q3. DIRECTIONS** *for question 3:* Select the correct alternative from the given choices.

What best can be said about the student who was the fastest to finish the first two laps of the race?

- ☐ a) It is Gaurav.
- ☐ b) It is Himesh or Gaurav.
- ☒ c) It is Himesh. ✓ **Your answer is correct**
- ☐ d) It is Gaurav or Jitesh.

**Q4. DIRECTIONS** *for question 4:* Type in your answer in the input box provided below the question.

What is the distance covered (rounded to the nearest meter) by the runner-up in the fourth lap before he was eliminated?

**Q5. DIRECTIONS** *for question 5:* Select the correct alternative from the given choices.

What is the total duration of the race?

- ☐ a) 660 seconds
- ☐ b) 600 seconds
- ☐ c) 540 seconds
- ☒ d) 720 seconds **✗ Your answer is incorrect**

**DIRECTIONS** for questions 6 to 10: Answer the questions on the basis of the information given below.

Five friends, Kalyan, Lohit, Ratan, Hari and Pavan, entered a supermarket, not necessarily at the same time, starting from 10:00 AM. They came out of the supermarket such that the last person among the five left the supermarket not later than 3:00 PM. Further, the amount that each person spent at the supermarket is a positive integral multiple of 100.

For any person,  $i$ , if the amount (in ₹) spent by him at the supermarket is  $A_i$  and the time (in minutes) that he spent at the supermarket is  $T_i$ , it is known that  $T_i$  was not more than half of  $A_i$  and was not less than one-third of  $A_i$ .

**Q6. DIRECTIONS** for questions 6 and 7: Select the correct alternative from the given choices.

If the total amount that the five persons spent at the supermarket was ₹700, what is the maximum time (in minutes) that any of the five persons could have spent at the supermarket?

- ☐ a) 150
- ☐ b) 200
- ☐ c) 100
- ☐ d) 300

**Q7. DIRECTIONS** *for questions 6 and 7:* Select the correct alternative from the given choices.

If four of the five persons spent the same amount in the supermarket and the amount that Pavan spent was two hundred more than that spent by Lohit and was twice that spent by Kalyan, what is the minimum time (in minutes) for which Pavan could have been in the supermarket?

- ☐ a) 50
- ☐ b) 100
- ☐ c) 200
- ☐ d) 133.33



**Q8. DIRECTIONS** *for questions 8 and 9:* Type in your answer in the input box provided below the question.

It is known that, between 10:00 AM and 3:00 PM, at any point in time, there was at most one person in the supermarket.

If amounts (in ₹) spent by Kalyan, Lohit and Ratan were in the ratio 1:2:3, what is the maximum amount (in ₹) that Hari could have spent in the supermarket?

**Q9. DIRECTIONS** *for questions 8 and 9:* Type in your answer in the input box provided below the question.

It is known that, between 10:00 AM and 3:00 PM, at any point in time, there was at most one person in the supermarket.

What is the maximum amount (in ₹) that any person could have spent in the supermarket?

**Q10. DIRECTIONS** *for question 10:* Select the correct alternative from the given choices.

If no two persons spent the same amount at the supermarket, what is the maximum possible duration (in minutes) for which all the five persons could have been in the supermarket together?

- ☐ a) 200
- ☐ b) 300
- ☐ c) 266.67
- ☐ d) 250

**DIRECTIONS** for questions 11 to 15: Answer the questions on the basis of the information given below.

Abhinav teaches Operations Management in Well-known Institute of Management in Western India. For each batch of students that he teaches, Abhinav, conducts exactly six tests, Test 1 through Test 6, each for a maximum of 100 marks. For awarding the final grade to each student, he considers the four highest marks scored by that student among the six tests. The sum of the four highest marks of each student is called the Applicable Score of that student.

The following table provides partial information about the marks scored by the seven students in the six tests and their Applicable Scores:

| Student | Test 1 | Test 2 | Test 3 | Test 4 | Test 5 | Test 6 | Applicable Score |
|---------|--------|--------|--------|--------|--------|--------|------------------|
| Drew    | 52     | 48     |        | 33     | 66     | 45     |                  |
| John    | 92     | 70     | 54     | 71     |        | 23     | 321              |
| Kevin   | 65     | 70     | 54     |        | 47     | 74     |                  |
| Phil    |        |        | 75     | 65     | 15     | 14     | 256              |
| Paul    | 54     | 78     | 77     |        | 58     | 46     | 267              |
| Jack    | 41     | 49     |        | 44     | 75     | 48     | 240              |
| Gary    | 62     | 75     |        | 71     | 38     | 56     |                  |

It is known that no two students obtained their four highest marks in the same set of four tests. Further, no student received the same mark across any two tests.

**Q11. DIRECTIONS** for questions 11 and 12: Select the correct alternative from the given choices.

What best can be said about the Applicable Score of Gary?

- ☐ a) It lies between 260 and 265.
- ☐ b) It is at least 265.
- ☐ c) It lies between 275 and 280.
- ☐ d) It is at most 260.

**Q12. DIRECTIONS** *for questions 11 and 12:* Select the correct alternative from the given choices.  
Among the given students, which student has the fourth highest Applicable Score?

- ☐ a) **Gary**
- ☐ b) **Paul**
- ☐ c) **Kevin**
- ☐ d) **Drew**

**Q13. DIRECTIONS** *for question 13:* Type in your answer in the input box provided below the question.

Among the seven students, what is the difference between the highest Applicable Score and the lowest Applicable Score?

**Q14. DIRECTIONS** *for questions 14 and 15:* Select the correct alternative from the given choices.

The marks of which test were considered for calculating the Applicable Score of the least number of students?

- ☐ a) **Test 3**
- ☐ b) **Test 4**
- ☐ c) **Test 5**
- ☐ d) **Test 6**

**Q15. DIRECTIONS** *for questions 14 and 15:* Select the correct alternative from the given choices.

For how many tests can the total marks scored by all the seven students in the test be determined accurately?

☐ a) 1

☐ b) 2

☐ c) 3

☐ d) 4



**DIRECTIONS** *for questions 16 to 20: Answer the questions on the basis of the information given below.*

Raghu, a salesman in a mobile phone manufacturing company, stays in Metropolis. He has to visit dealers in one or more cities among eight different cities – A through H. The distances (in km) of cities A through H from Metropolis are 200, 300, 250, 150, 100, 400, 350 and 200 respectively. After visiting any city, he must return to Metropolis (the distance from Metropolis to any city is same as the distance from that city to Metropolis).

During the next month, Raghu has to visit at least three cities, but he plans to travel for a total distance of at most 2000 km.

It is known that

- i. between C, E, G and H, he has to visit at least one city.
- ii. between A, B, D and F, he has to visit at least two cities.
- iii. between A, D, E and H, he cannot visit more than two cities.
- iv. if he visits C, he has to visit G.
- v. if he visits E, he has to visit A and vice versa.
- vi. if he visits B, he cannot visit C.

**Q16. DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.

What is the minimum distance that Raghu must travel in the next month?

☒ a) 900 km **✗ Your answer is incorrect**

☐ b) 1300 km

☐ c) 1200 km

☐ d) 1000 km

**Q17. DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.

What is the maximum number of cities that Raghu can visit in the next month?

☐ a) 3

☐ b) 4

☐ c) 5

☐ d) 6

**Q18. DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.

Which of the following cities will he definitely not visit?

☐ a) **F**

☐ b) **B**

☐ c) **C**

☐ d) **D**

**Q19. DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.

If Raghu selected the cities to travel such that he has to travel a total of exactly 1500 km, which of the following cities would he have selected?

☐ a) **A**

☐ b) **B**

☐ c) **G**

☐ d) **H**

**Q20. DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.

If Raghu cannot visit B unless he visits D, what is the minimum distance that Raghu must travel in the next month?

- ☐ a) 1300 km
- ☐ b) 1200 km
- ☐ c) 1800 km
- ☐ d) 1500 km